Oppg 1/

X = bloddtrykk før medisin

Y = bloddtrykk etter medisin

$D = Y - X$, $D \sim N(\mu, \sigma^2)$

Påstand: medisin er bloddtrykksnedsettende $= H_1$

$H_0: \mu = 0$ $H_1: \mu < 0$ $(y < x)$

c)

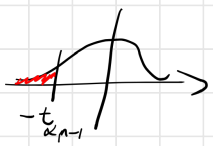
Testobservator: $T = \frac{\bar{D}}{s/\sqrt{n}}$, $\bar{D} = \frac{1}{n}\sum_i D_i$, $s^2 = \frac{1}{n-1}\sum_i (D_i - \bar{D}_i)$

observerte verdier: $n = 11$, $\bar{d} = -1,25$, $s = 3,2$

$\alpha = 0,05$

$T \sim t_{n-1}$ $\Rightarrow$ $t_{\alpha, n-1} = 1.812$

$H_1: \mu < 0$: $T \leq k$ ~>



kritisk verdi: $\underline{-1,812}$

d)
$$T_{obs} = \frac{-1.25}{3.2/\sqrt{11}} = -1.2956$$

e) teststyrke ≥ 0.8

+ Dc trer at H_1 = sann

$\Rightarrow P(\text{type II feil}) < 0.2$

Antar: $\mu = 0$ (men ukjent)
$\sigma^2 = s^2$ (kjent)

$$\left.\begin{array}{l}\text{Antar: } \mu = 0 \text{ (men ukjent)} \\ \sigma^2 = s^2 \text{ (kjent)}\end{array}\right\} Z = \frac{\bar{D} - \mu}{s/\sqrt{n}} \sim N(0,1), \quad \mu = 0$$

$P(\text{Type II feil}) = P(Z < z_\beta)$, $\beta = 0.05 \Rightarrow z_\beta = 1.645$

$= P(\bar{D} < z_\beta \cdot s/\sqrt{n})$

$= \Phi(z_\beta \cdot s/\sqrt{n}) < 0.2$

$z_\beta \cdot s/\sqrt{n} < z_\varphi$, $\varphi = 0.2 \Rightarrow z_\varphi = 0.842$

$$n > \left(\frac{z_\beta}{z_\varphi} \cdot s\right)^2 = \left(\frac{1.645}{0.842} \cdot 3.2\right)^2$$

$n > 39{,}085$ runde opp

$n \geq 40$

Oppg 2/

$Y \sim N(\mu, \sigma^2)$, μ = ukjent , σ = ukjent

$H_0 : \sigma \leq \sigma_0$ $H_1 : \sigma > \sigma_0$

Forkaster H_0 dersom $\frac{s^2(n-1)}{\sigma_0^2} > \chi^2_{n-1,\alpha}$

$n = 14$, $\alpha = 0.005$, $\sigma_0 = 2$

a) $29,819 = \chi^2_{\alpha, n-1}$

b) Teststyrke $\geq 95\%$ $\longrightarrow$ $P(\text{Type II, men } \sigma = \text{ukjent}) < 0.05$

$= P(\text{Forkast } H_0 \mid H_0 = \text{sann}) \geq 0.95$

Under H_0 testes σ_0

Under H_1 testes σ, trenger "Under H_0" som funk av σ

$H_0 = \text{sann}$: $\frac{s^2(n-1)}{\sigma_0^2} \sim \chi^2_{n-1,\alpha}$

$H_1 = \text{sann}$: $\frac{s^2(n-1)}{\sigma^2} \sim \chi^2_{n-1,1-\alpha} \longrightarrow s^2 = \frac{\sigma^2 \cdot \chi^2_{n-1}}{n-1}$

$$\frac{s^2(n-1)}{\sigma_0} = \frac{\sigma^2}{\sigma_0^2} \cdot \frac{\chi^2_{n-1}}{n-1} \cdot (n-1) = \frac{\sigma^2}{\sigma_0^2} \cdot \chi^2_{n-1,1-\alpha}$$

$$P(\text{Forkast } H_0 \mid H_0 = \text{sann}) = P\left(\frac{s^2(n-1)}{\sigma_0} > 29.819\right)$$

$$= P\left(\frac{\sigma^2}{\sigma_0^2}\chi^2_{n-1,1-\alpha} > 29.819\right) \geq 0{,}95$$

$$\Rightarrow \sigma^2 > \frac{29.819}{\chi^2_{n-1,0.95}} \cdot \sigma_0^2 = \frac{29.819}{5.892} \cdot 4$$

$$\underline{\sigma > 4{,}4993}$$

alternativt:

$P(\text{Type II feil}) \leq 0.05 = \beta$

$P(\text{ikke forkaste } H_0 \mid H_1 \text{ sann})$

$$P\left(\frac{s^2(n-1)}{\sigma_0^2} < \chi^2_{n-1,\alpha} \mid \sigma\right)$$

$$P\left(\underbrace{\frac{s^2(n-1)}{\sigma^2}}_{\sim \chi^2_{n-1,\beta}} < \chi^2_{n-1,\alpha} \cdot \frac{\sigma_0^2}{\sigma^2} \mid \sigma\right)$$

$$\chi^2_{0.95,n-1} < \chi^2_{0.005,n-1} \cdot \frac{\sigma_0^2}{\sigma^2} \Rightarrow \sigma^2 > \frac{29.819}{5.892} \cdot 2^2 = \underline{4{,}499}$$

Oppg 3/

$X \sim N(\mu, \sigma^2)$ μ = ukjent, $\sigma^2 = 1$

$H_0: \mu = 0$ mot $H_1: \mu > 0$ Testobservator: $T = \frac{\bar{X}}{\sigma/\sqrt{n}} = \bar{X}\sqrt{n}$

$\alpha = 0.05$, $n = 14$, $\bar{x} = 0.4$

a) $t_{obs} = \bar{x} \cdot \sqrt{14} = \underline{1{,}4967}$

b) $T \sim N(0,1) \Rightarrow t_\alpha = \underline{1.645}$

c) p-verdi $= P(T \geq t_{obs} \mid H_0 \text{ sann})$

$= P(T \geq 1{,}4967 \mid \mu = 0)$

$= 1 - \Phi(1{,}4967) = 1 - 0{,}9319 = \underline{0{,}0681}$

d) $P(\text{Forkast } H_0 \mid H_1 \text{ sann}, \mu = 0{,}84)$

$P(T > t_\alpha \mid \mu = 0{,}84)$

$= P\left(\frac{T - T_{forventet}}{\sqrt{\sigma}} > \frac{t_\alpha - T_{forventet}}{\sqrt{\sigma}}\right)$

$T_{forventet} = \mu\sqrt{n} = 0.84 \cdot \sqrt{14}$

$\Rightarrow 1 - \Phi\left(1.645 - 0.84\sqrt{14}\right)$

$= 1 - \Phi(-1{,}4980) = \underline{0{,}932}$

$= 1 - \beta$

$= 1 - P(\text{Type II feil})$

$= 1 - \text{p-verdi}$

$= \underline{0{,}932}$

* Dette er bare en tilfeldighet, pga omstendigheter

e) Fortegnstest, Testobservator: $U = \sum I(x_i > 0)$

forkaster H_0 dersom $U \geq k$

U_{obs}: 10

f) $U \sim binom(n, p)$, $n=14$, $\mu=0.84$, $\alpha=0.05$

Forkaster H_0 om $U \geq k$ der k bestemmes av:

$P(U \geq k \mid \mu=0) = (\leq) \alpha$

$1 - P(U < k \mid p=0.5) \leq \alpha$

$P(U < k \mid p=0.5) \geq 1-\alpha = 0.95$

$P(U<10)$ = første verdi med $P > 0.95$

$\Rightarrow 1-P(U<10) \Rightarrow \underline{\underline{P(U \geq 11)}}$ $\underline{k=11}$

g) $p = P(U \geq u_{obs} \mid p=0.5) = \sum_{u=10}^{n} \binom{n}{u} p^u (1-p)^{n-u}$ } python

$u_{obs}=10$, $n=14$

$\underline{\underline{p = 0.089783...}}$

h) Teststyrke : $1 - P(\text{Type II feil})$

$= P(\text{forkaster } H_0 \mid H_1)$

$= P(U \geq 11 \mid \mu=0.84)$

$= 1 - P(U < 10 \mid \mu=0.84)$

$\mu = 0.84 \Rightarrow P(X_i > 9) = ?$

$X \sim N(\mu, \sigma^2)$

$$P(X > 0) = \varphi$$

$$P\left(\frac{X-\mu}{\sigma} > \frac{-\mu}{\sigma}\right) = \varphi$$

$$1 - \Phi(-0,84) = 1 - 0,2005$$

$$= 0,7995 = p \quad \text{när } H_1 = \text{sann}$$

Teststyrka : $1 - P(U < 10 \mid p = 0,7995)$